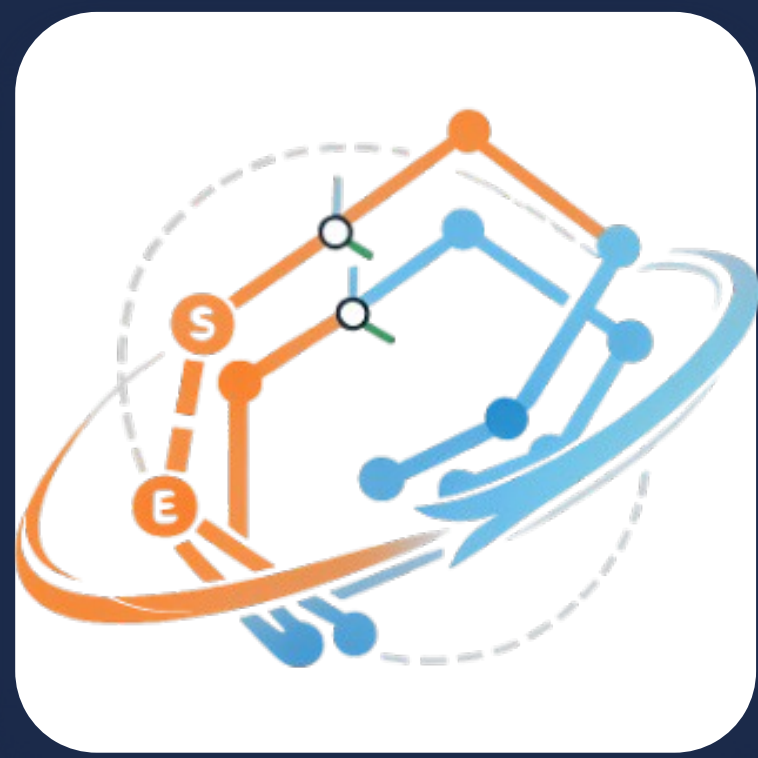




WARP



warp-retarget.github.io

Scan → project page

Whole-Body Retargeting for Learning from Offline Human Demonstrations

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TL;DR

WARP turns offline human demos into **precise, consistent whole-body robot actions in closed form** — open-loop replay & policy learning, no teleoperation in the loop.

ABSTRACT

Direct transfer from human demonstration to learnable robot action is key to scalable whole-body mobile manipulation. Human data scales better than mobile teleoperation but carries large embodiment gaps; existing retargeters are imprecise or inconsistent, creating action multi-modality that stops supervised policies from converging. WARP explicitly models embodiment differences to extract **precise, unique whole-body actions** via a closed-form Shoulder–Elbow–Wrist solver with lazy mobile-base control.

1

Why offline retargeting?

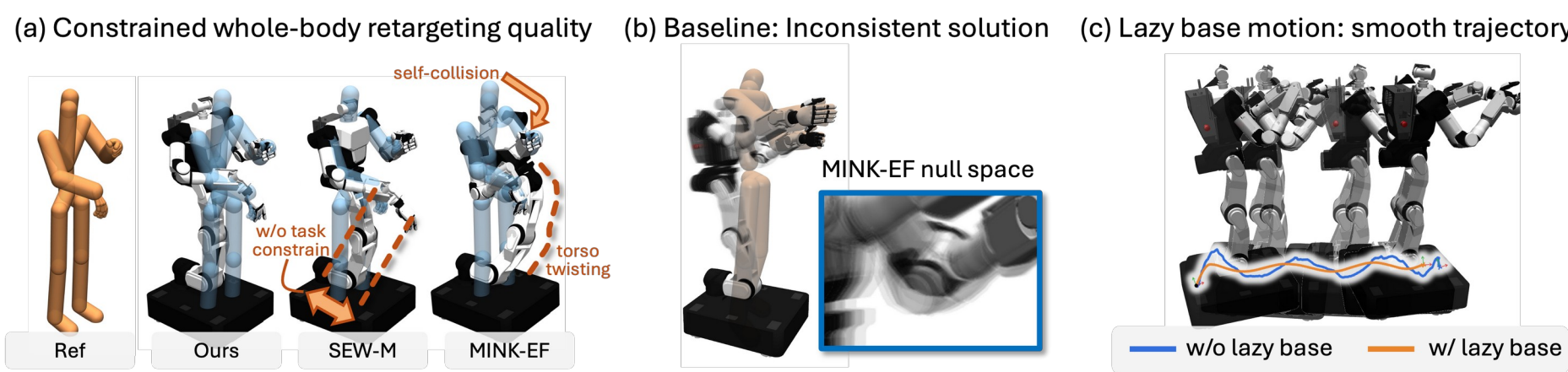
In teleoperation an operator absorbs error live. Offline there is none — the retargeted trajectory **is the supervision**.

FAILURE MODE 1 · IMPRECISION

Weighted-IK objectives trade end-effector accuracy against whole-body similarity. Either way, retargeting error becomes supervision error.

FAILURE MODE 2 · INCONSISTENCY

Redundant humanoids map nearly identical human poses to different robot configurations across solver seeds — similar observations, divergent actions.



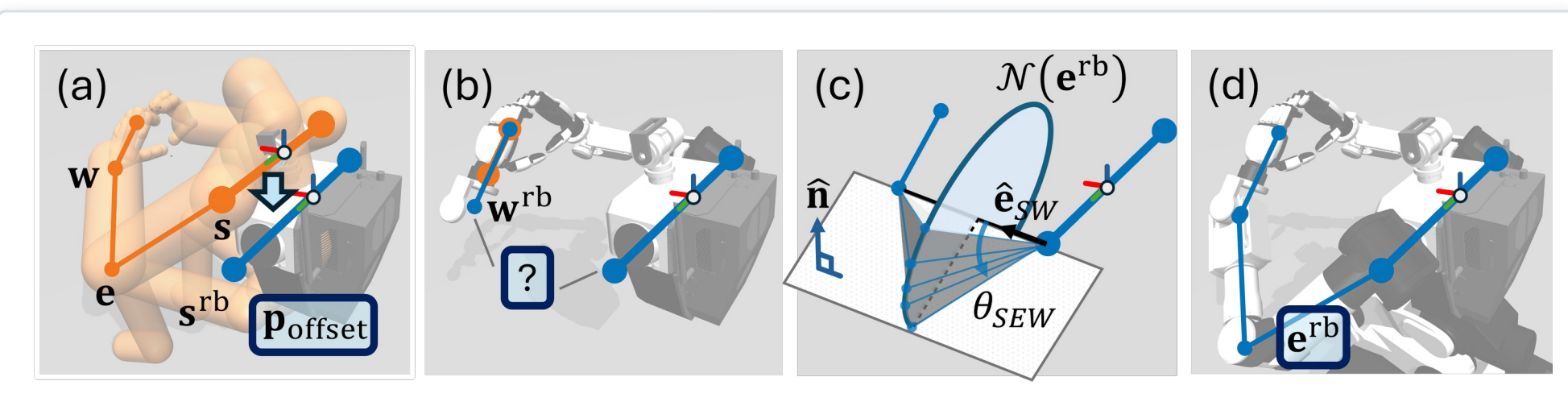
(a) match EEF and pose (b) closed form never varies (c) lazy base stays smooth

Key idea make retargeting a closed-form **function**, not an optimization — one exact, unique solution every frame.

2

Method: closed-form c-SEW

Solve retargeting in closed form over the Shoulder–Elbow–Wrist representation: the palm is matched exactly, remaining freedom preserves the human pose, joints follow analytically — **unique, exact, microseconds per frame, no calibration**.



(a) adaptive offset → (b) palm alignment → (c) elbow nullspace → (d) elbow angle resolves it

ADAPTIVE OFFSET

Shift the robot upper-body origin so palm centroids meet the human's — link-length gaps absorbed in closed form.

PALM ALIGNMENT

Hand orientation transfers directly; the wrist follows analytically and the elbow lands on the human's SEW half-plane.

LAZY MOBILE BASE

The 6-DoF torso absorbs small adjustments; the base moves only for genuine relocation — smooth base motion.

HIERARCHICAL POLICY

One flow-matching head with block-causal attention over base → torso → arm → hand — same inference cost.

Data single Meta Quest, whole body + hands @60 Hz, no mocap/robot in loop. **Robot** RB-Y1 — holonomic base, 6-DoF torso, 2×7-DoF arms, XHands @100 Hz.

3

Results

Retargeting quality

>150×

lower palm err
vs MINK-EF

≈0°

orientation err
(machine prec.)

~30×

faster solve
(hour vs day)

Fewest joint-limit & self-collision violations, **orders of magnitude more consistent** across seeds.

514 BONES-SEED-SOMA clips · vs MINK-EF, MINK-TE, SEW-M · lower is better.

Policy learning in simulation



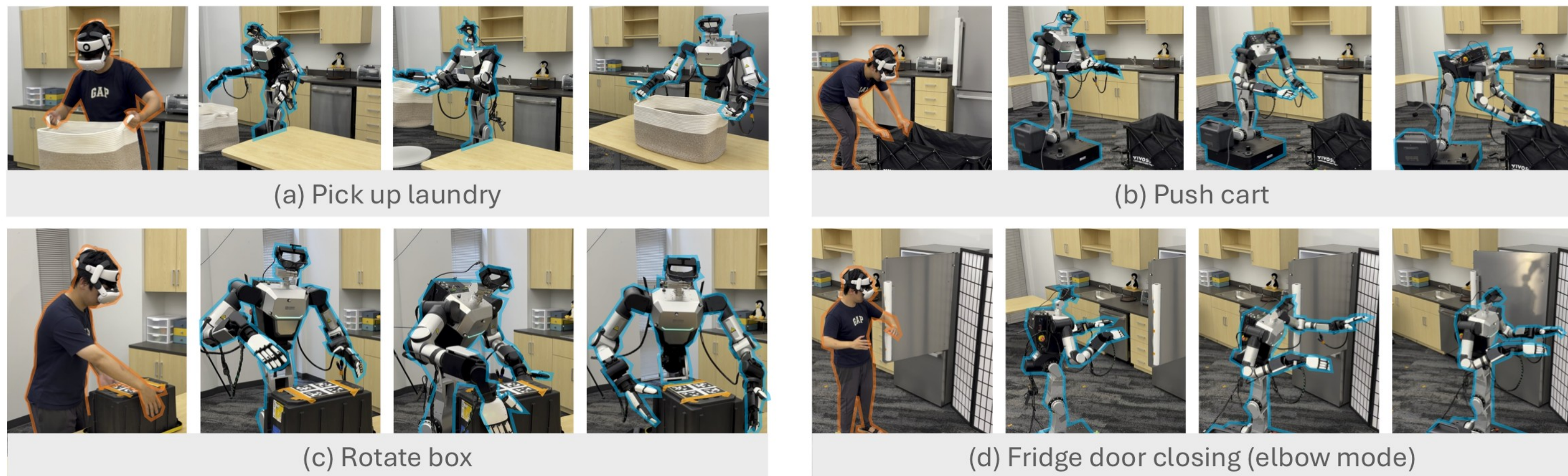
One human motion → many robot embodiments (Dual Kinova3, RB-Y1, OpenArm).

Policy rollout %	sort	pour	coffee	avg
MINK	94	74	8	59
WARP	100	78	34	71

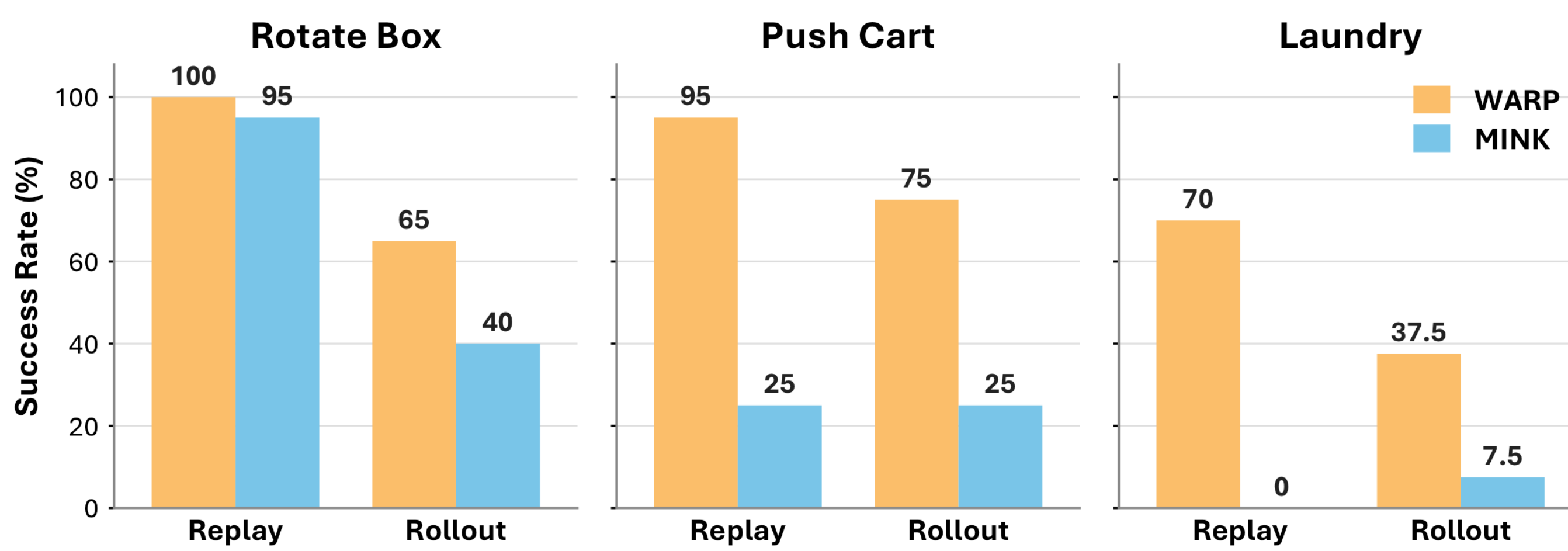
Replay comparable, but WARP-trained policies win **+12% on average** — nuances invisible at replay decide downstream success.

Real-world evaluation

RB-Y1 hardware · 50 human demos per task · WARP wins on all four



laundry (bimanual wrists) · cart (base–arm contact) · box (torso–arm twist) · fridge — closed with the elbow, which EEF-only methods cannot express.



Replay & policy-rollout success (%), 10 trials each.

90% replay on the elbow-contact fridge, and stronger policies even at equal replay (**65% vs 40%** on rotate-box) — motion quality, not replay completion, governs downstream success.

First zero-shot whole-body mobile manipulation directly from offline human demos — no human-in-the-loop teleoperation.

Robotics: Science and Systems (RSS) 2026 · Poster